PROJECT-BASED LEARNING

PHASE-I EVALUATION

Detective Mystery: The Hidden Truth

Team ID: DSCPP-III-2026-T152

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: T152
Semester: 3rd
Section: DS2,F,J,K
Domain: *Game Development*

Team Members

1. *Karan Badhani : 2510360037*
2. *Khushi Singal : 2510360039*
3. *Suraj Giri Goswami : 2510012585*
4. *Tanmay Arora : 2510610178*

Mentor

Prof. Dr. Jyoti Agarwal

Interactions completed: 3

Problem Statement & Motivation

Problem Statement

Most students practice OOP and DSA through separate programs, so it becomes difficult to see how these concepts work together in one application. This project presents a C++ detective mystery game where the player investigates clues, evidence and suspects while the game logic applies OOP and DSA concepts.

Motivation

- *Connect C++ concepts with a playable project*
- *Build logical thinking through investigation*
- *Use DSA inside actual game actions*

Target Users / Stakeholders

- *Primary users: Students and game players*
- *Secondary stakeholders: Teachers and faculty*
- *Expected benefit: Better understanding of OOP and DSA*

Objectives

Project Objectives

1. *Create a 2D detective mystery game using C++.*
2. *Add gameplay features such as location exploration, clue collection, suspect interaction, evidence checking and final deduction.*
3. *Organize the game using OOP concepts such as classes, objects, encapsulation, inheritance and polymorphism.*
4. *Use DSA concepts for map navigation, investigation records, clue searching and evidence arrangement.*
5. *Test the prototype through basic cases for movement, clue handling, investigation flow and final verdict.*

Key Objective: *Build a detective game where the player solves a case while the project demonstrates practical use of C++ OOP and DSA.*

Proposed Solution

Proposed Approach

1. *Create the case and player setup*
2. *Allow movement through connected locations*
3. *Collect clues and question suspects*
4. *Use DSA for investigation handling*
5. *Show verdict and score*

Key Features

- *Playable mystery investigation*
- *Graph-based location movement*
- *Clue and evidence storage*
- *Suspect interaction system*
- *Final deduction with scoring*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Arrays, Linked Lists, Trees, Graphs, Hashing, Searching, Sorting*	*Investigation and clue management*
OOPs with C++	*Classes, Objects, Inheritance, Polymorphism, Encapsulation*	*Game entities and investigation engine*
Operating Systems	–	–
DBMS	–	–

Requirement: OOP will be used to create and manage the main game entities, while DSA will support navigation, clue handling, investigation history and evidence analysis.

System Architecture / Workflow

Input
- Player Actions
- Case Data
- Clues / Evidence
- Locations

→

Processing Layer
- Check Input
- Update Game State
- Handle Events
- Apply Rules

→

Core Logic
- Investigation Flow
- Clue Matching
- Suspect Analysis
- DSA Operations

→

Data Layer
- Player Data
- Case Details
- Clues / Evidence
- Suspects
- Locations
- Progress

→

Output / Result
- Case Progress
- Final Deduction
- Verdict
- Score

Major Components

- *Player and game management*
- *Location and investigation management*
- *Clue, evidence and suspect management*

Data / Control Flow

- *Player gives input and case data is loaded*
- *Game logic processes clues and evidence*
- *Progress is updated after each action*
- *Final output shows verdict and score*

Technology Stack & Methodology

Technology Stack

- Programming: *C++*
- Data Storage: *File handling*
- Tools / Framework: *SFML*
- IDE: *Visual Studio Code*
- Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

C++ supports OOP and DSA implementation, while SFML can help in creating a simple 2D game interface.

Initial Progress & Team Contribution

Progress Achieved

- *Basic idea and outline of the mystery prepared*
- *Main characters, locations, clues and evidence planned*
- *Initial project requirements and game flow discussed*
- *Basic system design and project structure prepared*

Individual Contribution

Member	Role	Contribution
M1	*Lead*	*Storyline, planning and system design*
M2	*Developer*	*Player movement and game mechanics*
M3	*DSA Developer*	*DSA integration and investigation logic*
M4	*Documentation/UI*	*Documentation, UI and testing*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal and Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

- *Working 2D detective game prototype*
- *Practical use of OOP and DSA in gameplay*
- *Scope to add more cases and levels later*

References

1. *SFML Official Documentation*
2. *C++ Reference*
3. *GeeksforGeeks – Data Structures*
4. *Microsoft Learn – C++ Documentation*

Thank You
Questions & Suggestions